CRIZ (Admin IQ)

Budget Estimate Summary

Laabu AI Workforce · Phase 3 – Budget

Item	Details
Project	Laabu AI Workforce
Agent	CRIZ (Admin IQ)
Document Type	Budget Estimate Summary
Lifecycle Phase	Phase 3 – Budget
Status	Draft for Review
Deployment Target	Google Cloud First, Cloud-Neutral Core

1. Executive Summary

This document provides planning-level cost estimates for CRIZ (Admin IQ), the operations backbone of the Laabu AI Workforce. All figures are estimates based on current cloud pricing and the workload assumptions described in the Planning and Design documents. Actual costs will be confirmed through the pilot and reported in the Cost Report.

CRIZ operates in two modes: an External Partner and Vendor Concierge, and an Internal Operations and Workflow Assistant. It supports nine operational capability modules and runs on Google Cloud Platform for Version 1, with a cloud-neutral core designed for future AWS Bedrock and Azure AI Foundry support.

The budget covers three cost areas: development and setup, monthly platform operation during the pilot, and cost-per-workflow estimates for ongoing use. All estimates are presented at planning accuracy (+/- 30%) and must be reviewed after the pilot before a production commitment is made.

Cost Area	Estimate	Notes
Development and Setup (one-time)	**USD 18,000 – 26,000**	Includes build sprints, testing, and GCP setup.
Monthly Platform Cost – Pilot	**USD 600 – 1,100 / month**	Low to medium vendor and operations volume.
Monthly Platform Cost – Production	**USD 1,800 – 3,800 / month**	Full nine-module deployment at target volume.
Estimated Cost per Workflow	**USD 0.08 – 0.55**	Varies by capability and model level used.
Contingency Reserve (15%)	**Included in sprint estimates**	Recommended for a first deployment of this scope.

All figures are planning estimates. Final costs depend on actual usage, model selection, tool call volume, and document processing activity.

2. What This Budget Covers

This estimate covers all costs required to design, build, test, and operate CRIZ Version 1 on Google Cloud Platform. It addresses both one-time development costs and recurring monthly operating costs during the pilot and production phases.

Area	What Is Included
Development and Setup	Sprint team time, GCP project configuration, MCP service setup, schema and interface design, fake tool builds for testing, and evaluation harness.
AI Model Usage	Vertex AI token costs for the workhorse model (routing, classification, drafting) and the reasoning model (compliance analysis, contract review, reporting).

Area	What Is Included
Workflow Orchestration	Vertex AI Agent Engine hosting and Cloud Run MCP service costs for all ten approved tools.
Document Processing	Cloud Storage reads and writes for vendor documents, contracts, and generated reports. Document AI usage through document-mcp.
Workflow State and Storage	Firestore reads and writes for nine operational collections including vendors, contracts, approvals, and audit events.
Approval and Audit Workflows	Approval service API calls, audit sink writes, and BigQuery query costs for operational reporting.
Security and Compliance	Secret Manager access, Sensitive Data Protection scans, and IAM policy management.
Monitoring and Observability	Cloud Logging, Cloud Trace, and Cloud Monitoring for workflow health, latency, cost, and error tracking.
Scheduling	Cloud Scheduler job creation and execution for reminders, follow-ups, and recurring operational tasks.
Testing and Evaluation	Evaluation run costs including agent evaluation prompts, contract tests, and security test execution.

3. Budget Target

The budget target covers two phases: the development phase ending with pilot go-live, and the ongoing monthly operating cost at pilot and production volume.

Phase	Target Range	Duration	Trigger to Move Forward
Phase 1 – Development and Setup	USD 18,000 – 26,000	10–14 weeks	Planning, Design, and Budget approved.
Phase 2 – Pilot Operation	USD 600 – 1,100 / month	3 months minimum	Pilot go-live approved. All deployment gates passed.
Phase 3 – Production Operation	USD 1,800 – 3,800 / month	Ongoing	Pilot evaluation passed. Actual cost report submitted.

Fixed versus variable costs are separated in this budget because the fixed platform costs continue regardless of usage, while variable costs scale with actual workflow volume.

Cost Type	Components	Pilot Estimate	Production Estimate
Fixed (platform base)	Agent Engine minimum, Cloud Run base instances, Monitoring, IAM, Secret Manager	USD 200 – 350 / month	USD 400 – 650 / month
Variable (usage-driven)	AI model tokens, Firestore reads/writes, document processing, tool calls, audit writes, BigQuery queries	USD 400 – 750 / month	USD 1,400 – 3,150 / month
Total		USD 600 – 1,100 / month	USD 1,800 – 3,800 / month

4. Main Workload Assumptions

All cost estimates are based on the following planning-level workload assumptions. These should be reviewed and confirmed with the operations team before the pilot begins.

Assumption	Pilot Volume	Production Volume	Notes
Vendor onboarding requests per month	10 – 20	50 – 150	Each request involves document collection and compliance review.
Contract lifecycle actions per month	10 – 30	60 – 200	Includes draft, review, signature routing, and archival.
Procurement requests per month	15 – 40	80 – 250	Includes PO creation and budget validation.
Travel requests per month	20 – 50	100 – 400	Includes policy check, approval, and booking confirmation.
Meeting coordination events per month	30 – 80	150 – 600	Includes scheduling, invites, minutes, and follow-ups.
Task tracking records per month	50 – 150	300 – 1,000	Includes creation, monitoring, escalation, and closure.
Operational report runs per month	5 – 15	20 – 80	May include BigQuery queries for larger date ranges.
Average documents processed per vendor onboarding	3 – 6	3 – 8	Compliance certificates, registration docs, and contracts.
Human approval actions required per month	20 – 60	100 – 400	Sensitive actions paused for explicit approval.

Assumption	Pilot Volume	Production Volume	Notes
Average workflow duration	2 – 8 minutes	2 – 8 minutes	Varies by capability and number of tool calls.

5. Model Cost Strategy

CRIZ uses two model capability levels as defined in the Design document. The workhorse model handles the majority of requests at lower cost. The reasoning model is reserved for complex decisions and is invoked only when required by the capability module rules.

Model names are not hard-coded. Changing the approved model requires a configuration change, evaluation, and a cost review. Token estimates below are based on current Vertex AI pricing for comparable model tiers.

Model Level	Used For	Estimated Input Tokens per Workflow	Estimated Output Tokens per Workflow	Estimated Cost per 1,000 Workflows
Workhorse	Routing, classification, draft communications, short summaries, field extraction, scheduling.	800 – 2,000	200 – 600	USD 15 – 45
Reasoning	Compliance analysis, contract review, multi-document processing, operational reports.	2,000 – 6,000	600 – 2,400	USD 80 – 280

The majority of CRIZ workflows will use the workhorse model only. The reasoning model is invoked in an estimated 15–25% of workflows — primarily vendor compliance reviews, contract analysis, and operational report generation.

Workflow Mix Estimate	Pilot (% of volume)	Production (% of volume)
Workhorse model only	75 – 85%	75 – 85%
Workhorse + Reasoning model	15 – 25%	15 – 25%
Estimated average blended cost per workflow	USD 0.08 – 0.30	USD 0.08 – 0.30

6. Main Cost Drivers

The following components account for the majority of monthly operating cost. Each driver is linked to specific CRIZ capabilities and can be monitored individually.

Cost Driver	Linked Capability	Pilot Monthly Estimate	Production Monthly Estimate	Variable?
AI model tokens (workhorse)	All nine modules	USD 80 – 180	USD 300 – 700	Yes
AI model tokens (reasoning)	Compliance, contracts, reporting	USD 40 – 120	USD 200 – 600	Yes
Vertex AI Agent Engine	CRIZ orchestrator runtime	USD 80 – 140	USD 160 – 280	Partly
Cloud Run (MCP services)	All ten tool services	USD 60 – 110	USD 120 – 220	Partly
Firestore reads and writes	All workflow state collections	USD 20 – 60	USD 100 – 300	Yes
Cloud Storage	Document, contract, and report storage	USD 10 – 30	USD 50 – 200	Yes
Document processing (document-mcp)	Vendor onboarding, contracts, AP	USD 30 – 80	USD 150 – 500	Yes
BigQuery queries	Operational reporting module	USD 10 – 30	USD 50 – 200	Yes
Secret Manager	All tool and model credentials	USD 5 – 10	USD 10 – 20	No
Cloud Logging and Monitoring	All workflows	USD 20 – 50	USD 60 – 150	Partly
Sensitive Data Protection scans	External document inputs	USD 15 – 40	USD 80 – 250	Yes
Cloud Scheduler	Reminders, follow-ups, recurring jobs	USD 5 – 10	USD 20 – 50	Yes
Audit event writes	All workflows (append-only)	USD 5 – 15	USD 30 – 100	Yes
IAM and networking	Platform baseline	USD 10 – 20	USD 20 – 40	No

Document processing is the highest-variable cost driver. Vendor onboarding workflows with multiple compliance documents will cost more per workflow than simpler task or meeting requests. BigQuery costs depend on the date range and data volume queried in operational reports.

7. Budget-Efficiency Controls

CRIZ is designed to track and control cost at the workflow level. The following controls must be active before the pilot begins.

Control	What It Does	How It Is Enforced
Spending alerts	Notify the operations team when monthly cost reaches 70% and 90% of the approved budget.	GCP Billing Alerts configured before pilot go-live.
Hard spend caps	Prevent GCP project spend from exceeding the approved monthly limit.	GCP Budget with hard cap enabled.
Cost labels on all resources	Tag every GCP resource with project, agent, capability, and environment labels.	Enforced in the deployment configuration. Unlabelled resources are flagged.
Per-workflow cost tracking	Record estimated token cost, tool call cost, and document processing cost for every workflow run.	Stored in the Telemetry interface; visible in the cost dashboard.
Model level routing rules	Route to the workhorse model by default. The reasoning model is invoked only when the capability module requires it.	Enforced in the orchestrator. No capability module may call the reasoning model without an explicit rule.
Token output limits	Cap maximum output tokens per model call by model level.	Set in configuration: workhorse 800 tokens, reasoning 2,400 tokens.
Document processing limits	Reject documents that exceed the approved file size or type limits before invoking document-mcp.	Checked in the orchestrator before any document tool call.
BigQuery byte limits	Restrict BigQuery queries to approved views with byte scan limits.	Applied through approved view definitions and IAM query permissions.
Pilot volume cap	Limit total workflow volume during the pilot to confirm cost estimates before scaling.	Operations team sets a monthly workflow limit in the pilot configuration.

8. Development Sprint Estimate

Development is structured into working sprints. Each sprint produces a testable, deployable capability. Sprint costs include engineering time and GCP development environment usage.

Sprint	Deliverable	Duration	Estimated Cost	GCP Dev Cost
Sprint 1	Project skeleton, typed schemas, interfaces, and fake tool layer.	1.5 weeks	USD 2,200 – 3,000	USD 50 – 100
Sprint 2	Vendor onboarding and compliance flow end-to-end.	2 weeks	USD 3,000 – 4,200	USD 80 – 150
Sprint 3	Contract lifecycle and e-signature workflow.	2 weeks	USD 2,800 – 3,800	USD 80 – 130

Sprint	Deliverable	Duration	Estimated Cost	GCP Dev Cost
Sprint 4	Procurement, travel, and meeting coordination modules.	2 weeks	USD 2,800 – 3,800	USD 80 – 130
Sprint 5	Task tracking, executive assistance, and operational reporting.	2 weeks	USD 2,600 – 3,600	USD 80 – 130
Sprint 6	GCP adapter connection, fake tool replacement, and integration tests.	1.5 weeks	USD 2,000 – 2,800	USD 100 – 200
Sprint 7	Security tests, agent evaluations, cost tracking, and runbook.	1.5 weeks	USD 1,800 – 2,400	USD 60 – 100
Contingency (15%)	Buffer for scope changes, unplanned complexity, and re-testing.		USD 2,730 – 3,570	
Total		~12–14 weeks	USD 20,000 – 27,000	USD 530 – 940

Sprint cost estimates assume a small, experienced team. Costs will vary based on team composition, hourly rates, and the number of parallel workstreams. GCP development environment costs are separate from and lower than the pilot operating costs.

9. Simple Scale View

The table below shows how estimated monthly operating costs change as workflow volume grows. These are planning-level projections based on the cost drivers in Section 6.

Monthly Workflow Volume	AI Model Cost	Infrastructure and Storage	Document Processing	Monitoring and Audit	Estimated Total
200 – 500 workflows (Pilot)	USD 120 – 300	USD 175 – 340	USD 45 – 110	USD 30 – 70	**USD 370 – 820**
500 – 1,500 workflows (Early Production)	USD 300 – 700	USD 280 – 500	USD 110 – 350	USD 60 – 130	**USD 750 – 1,680**
1,500 – 5,000 workflows (Full Production)	USD 700 – 2,000	USD 400 – 750	USD 350 – 1,100	USD 120 – 250	**USD 1,570 – 4,100**
5,000+ workflows (Scale)	USD 2,000 – 5,000+	USD 600 – 1,200+	USD 1,100 – 3,000+	USD 200 – 500+	**USD 3,900 – 9,700+**

Costs grow roughly linearly with workflow volume in the pilot and early production range. At scale, unit costs may decrease if GCP committed use discounts are applied. BigQuery costs for large operational reports can create spikes at any volume tier and should be monitored separately.

10. Cost and Business Value

CRIZ is designed to reduce manual operational effort, speed up vendor and partner workflows, and provide a complete and auditable record of every operational action. The table below summarises the expected operational benefits that justify the investment.

Capability	Current Manual Effort	Expected CRIZ Benefit	Measurable Outcome
Vendor Onboarding	3–5 days per vendor, manual document chasing.	Automated document collection and compliance check within hours.	Reduced time-to-active from days to hours.
Compliance Coordination	Manual policy comparison, spreadsheet tracking.	Automated policy gap detection with structured output.	Fewer missed compliance items; audit-ready records.
Contract Lifecycle	Email chains, manual signature tracking, version confusion.	Structured e-signature routing with status tracking and archival.	Faster contract completion; no lost document versions.
Procurement Management	Paper or email-based requests, manual approvals.	Automated policy checks, digital approval routing, PO issuance.	Faster approval cycles; spend policy compliance.
Travel Coordination	Manual policy checks, back-and-forth approvals.	Policy check, approval routing, and booking confirmation in one flow.	Reduced admin time; policy adherence enforced consistently.
Meeting Management	Manual calendar coordination, lost minutes.	Automated scheduling, agenda distribution, and follow-up creation.	Fewer scheduling conflicts; action items captured every time.
Task Tracking	Informal tracking, missed follow-ups.	Structured task assignment, monitoring, and escalation.	Nothing falls through the gaps; escalations happen on time.
Operational Reporting	Manual data pulls, inconsistent formats.	On-demand reports with cited sources and freshness timestamps.	Faster decisions; consistent and auditable reporting.

At a pilot cost of approximately USD 600 – 1,100 per month, CRIZ needs to save the equivalent of a few hours of operational staff time per week to justify its operating cost. The full business case will be confirmed in the pilot evaluation report.

11. Future AWS and Azure Reserve

The CRIZ design uses a cloud-neutral architecture. The application core communicates only through standard interfaces. Cloud-specific code is isolated in adapter folders. Version 1 implements GCP adapters only. AWS Bedrock and Azure AI Foundry support requires only new adapter implementations, not a rewrite of business logic.

The following reserve estimates cover the adapter build effort and any GCP-to-cloud migration testing costs, should a second cloud provider be required.

Future Workstream	Scope	Estimated Effort	Estimated Cost
AWS Bedrock adapter build	Implement ModelProvider, StateStore, ObjectStore, SecretStore, Telemetry, and Scheduler adapters for AWS.	3 – 4 weeks	USD 4,500 – 7,000
Azure AI Foundry adapter build	Implement the same interfaces for Azure AI Foundry, Cosmos DB, Blob Storage, and Key Vault.	3 – 4 weeks	USD 4,500 – 7,000
Portability validation tests	Run contract and portability tests for both new adapter sets against the approved interface standards.	1 week	USD 1,200 – 2,000
Cloud-neutral reserve (recommended)	Budget reserve to be held unspent until a second cloud provider is confirmed as a requirement.	Ongoing	USD 3,000 – 5,000

No AWS or Azure spend is planned in Version 1. The interfaces are designed and contract-tested in Version 1 at no additional infrastructure cost. The reserve estimate above applies only when a second cloud provider deployment is approved.

12. Costs Not Included

The following items are excluded from this budget estimate. They should be assessed separately and approved as part of any expansion scope.

Excluded Item	Reason for Exclusion
Engineering team salaries or agency contracts	Team composition and rate agreements are managed outside this budget.
ERP connector licence or subscription costs	First ERP connector is an open decision. Licence costs depend on provider selection.
E-signature provider subscription	First signature provider is an open decision. Licence costs depend on provider selection.

Excluded Item	Reason for Exclusion
Procurement system connector licence	Depends on the approved procurement platform. Licence costs are separate.
Data residency or regional compliance uplift	Regional deployment requirements are an open decision. Multi-region costs may apply.
Staff training and change management	Operational team onboarding and training are out of scope for this technical budget.
Legal review of compliance workflows	Legal sign-off on compliance logic and policy mapping is a business cost, not a platform cost.
GCP Marketplace or partner service fees	Third-party services accessed via GCP Marketplace are billed separately.
Disaster recovery or multi-region failover	Business continuity architecture is not in scope for Version 1.
Penetration testing by an external firm	External security testing is a separate procurement. Internal security tests are included.

13. Actual Cost Tracking

Cost tracking is a first-class requirement for CRIZ, not an afterthought. The following metrics must be recorded for every workflow run and reported in the monthly cost dashboard.

13.1 Per-Workflow Metrics

Metric	What Is Tracked	Where It Is Stored
workflow_id	Unique identifier for every workflow run.	Telemetry and Firestore audit_events.
capability_module	Which of the nine CRIZ modules handled the request.	Telemetry and Firestore audit_events.
model_level	Workhorse or reasoning model used.	Telemetry and Firestore audit_events.
input_tokens	Total input tokens consumed by all model calls in the workflow.	Telemetry per-workflow record.
output_tokens	Total output tokens produced by all model calls.	Telemetry per-workflow record.
tool_calls	Number and type of tool calls made (email, document, signature, etc.).	Telemetry per-workflow record.
document_count	Number of documents processed through document-mcp.	Telemetry per-workflow record.
approval_required	Whether the workflow required a human approval pause.	Telemetry and approval collection.

Metric	What Is Tracked	Where It Is Stored
end_to_end_latency_ms	Total elapsed time from request received to result returned.	Telemetry per-workflow record.
estimated_cost_usd	Estimated USD cost for the workflow based on token and tool usage.	Telemetry per-workflow record.
outcome	Success, failed, paused for approval, or routed to human review.	Telemetry and audit_events.

13.2 Monthly Dashboard Metrics

Dashboard Metric	Target
Total workflows processed	Track month-on-month. Flag if volume exceeds the pilot cap.
Cost per workflow by capability	Monitor for outlier capabilities driving disproportionate cost.
Model token spend (workhorse vs reasoning)	Reasoning should remain below 30% of total token spend.
Document processing cost as % of total	Flag if document cost exceeds 35% of monthly total.
Workflow success rate	Target: at least 95%.
Human approval rate	Track what percentage of workflows pause for approval.
Duplicate prevention events	Should remain at zero for any production workflow.
Average end-to-end latency	Target: under 8 seconds for workhorse workflows; under 20 seconds for reasoning workflows.
Cost forecast vs approved budget	Alert at 70% and 90% of monthly budget. Hard cap at 100%.

14. Budget Review Checklist

The following checklist must be completed and signed off before CRIZ is approved for production deployment. All items must have a confirmed status.

#	Checklist Item	Owner	Status
1	Planning, Design, and Budget documents are approved by the project sponsor.	Project Lead	Pending
2	Workload assumptions in Section 4 have been reviewed and confirmed by the operations team.	Operations Lead	Pending

#	Checklist Item	Owner	Status
3	GCP project billing account is linked and the approved monthly budget cap is set.	Platform Engineer	Pending
4	GCP Budget Alerts at 70% and 90% of the monthly cap are configured and tested.	Platform Engineer	Pending
5	All GCP resources have the required cost labels (project, agent, capability, environment).	Platform Engineer	Pending
6	Per-workflow cost tracking is active and visible in the cost dashboard.	Platform Engineer	Pending
7	The pilot volume cap is set in the CRIZ configuration.	Operations Lead	Pending
8	Model token output limits are set in the approved configuration (workhorse: 800, reasoning: 2,400).	Platform Engineer	Pending
9	BigQuery approved views and byte scan limits are in place.	Data Engineer	Pending
10	Document processing file size and type limits are enforced in the orchestrator.	Platform Engineer	Pending
11	All deployment gates in Section 23 of the Design document have passed.	QA Lead	Pending
12	Security scan has no blocking findings.	Security Lead	Pending
13	Agent evaluations meet all required pass targets (intent routing 95%+, sensitive data removal 100%, approval block 100%).	QA Lead	Pending
14	Actual cost during the pilot phase is compared to the estimate in this document.	Finance Lead	Ongoing (Pilot)
15	Cost Report is submitted after the pilot before production approval is granted.	Project Lead	After Pilot

This checklist should be reviewed at the end of each sprint and completed in full before the pilot go-live gate. Any item marked as not confirmed must be escalated to the project sponsor before production deployment is approved.